

Biofuels

Biofuels have caused much strife because of the rise in food prices after lands were diverted to produce them. India, however, cannot turn its back on them. Nine states have already passed laws requiring blending of gasoline with upto 5 per cent ethanol from sugarcane. Production of biodiesel from *Jatropha curcas* and *Pongamia* shrubs has increased. Now 26 states are establishing plantations to grow the crops. But its quality needs to be standardised as oil content tends to vary. Experts argue that biodiesel would be more efficient for household cooking instead. The major constraints of these fuels are their high costs of production.

Save Paper, Use G-mail



Instead of penning down your thoughts or letters on reams of paper made from felling trees, try G (green) mail instead. Okay, it's a cheap pun but at least you get the message. Offices all across the world are telling their employees to cut down on printing or faxing messages and instead using e-mail. You would be surprised by just how many trees you can save from doing so. Which then become major sinks for carbon emissions.

Shut the Windows

Properly caulking and fastening windows and even doors can reduce heating and cooling usage by as much as 20 per cent. Specially-coated windows are now available that improve heat or cold retention. Alternatively, for those who don't want run their air conditioners in summer, the advice is, open your windows wide and let the air in. But don't try it during a Delhi summer.

In the Genes

Application of biotechnology to agriculture holds the key to drought-proof seeds, higher yields despite higher CO₂ concentration and protection from stubborn pests. India must make major investments in agricultural research to boost the expected fall in productivity of crops because of higher thermal stresses.

Not Stainless

Iron and steel plants contribute more than half of India's carbon emissions from the industrial sector. There is an immediate need to upgrade manufacturing technology in most steel plants across the country. State-of-the-art emission control technology needs to be introduced and governments could support such ventures.

Small is Beautiful

With large dams facing resistance from eco-warriors, it's time India looked seriously at exploiting small-scale hydropower projects of upto 25 MW. Such projects can generate 15,000 MW power, of which only around 2,000 MW has been utilised. These plants are competitive with conventional options for rural electrification, especially in remote areas.



Black Carbon Worry

Black carbon emissions are something experts increasingly worry about. In India and China, it's because of the burning of wood and cow dung for household cooking and is believed to cause the Asian brown cloud—a swathe of polluted air. Better cooking stoves are a way out but the long term answer would be to switch to modern fuels such as LPG for cooking.

The Agri-crisis

Production of cereals, like wheat, is expected to fall because of climate change. The Government has identified sustainable agriculture as a national mission and intends to focus on four areas: dryland agriculture, risk management, the use of bio-technology and access to information.

YASBANT NEGI



Food security is a major issue in India and the country needs to move fast to stave off the crisis.